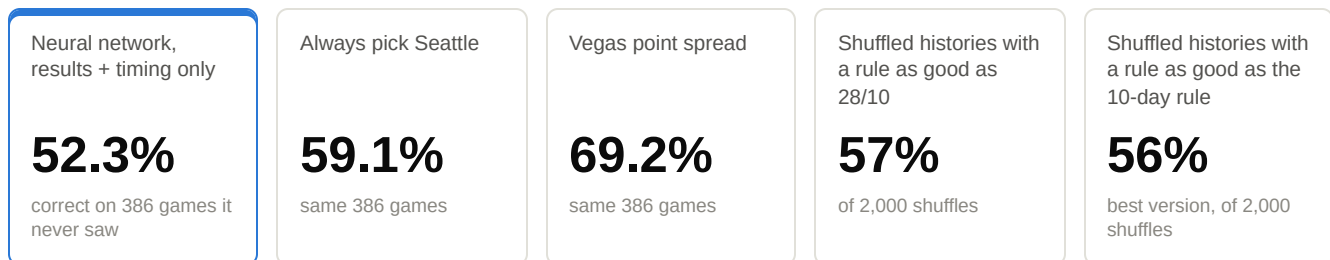


Do Seattle's wins and losses follow the calendar?

A neural network and a set of statistical tests, applied to every Seattle game since 1999, to the 28/10-day rule, and to the 10-day, 4th-previous rule.

Prepared Sep 28, 2026 · Game data through Sep 27, 2026 (at WAS, L 31–33) · Source: nflverse games data · Reproducible with `node cli.js report`



Summary

1. **The neural network found nothing usable.** Given only Seattle's own results and the calendar spacing of its games, it picked 52.3% of 386 unseen games (2004–2025) correctly, significantly worse than simply picking Seattle every week (59.1%; McNemar $p = 0.03$). The Vegas point spread got 69.2%.
2. **The real order of wins and losses is not special.** When each season's results were dealt out in random order 200 times, the same network did as well on the shuffled histories (53.5% on average) as on the real one ($p = 0.63$).
3. **Both rules' records are real, and both are what searching produces.** The 28/10 rule is 8-for-8 and the 10-day, 4th-previous rule 9-for-10 in 1999–2025 (9-for-9 if all games must be in one season). After accounting for the 202 similar rules examined, a record at least this good turns up in 56%–95% of shuffled histories.
4. **Both rules lean on two ordinary facts:** good seasons stay good (so “repeat the result” beats 50%), and after a Thursday game Seattle happened to repeat the Thursday result 11 of 14 times. For the other 31 teams the same rules are right about 60% of the time, as shuffled seasons predict.
5. **The numbers 3, 4, 7, 10, 11 and 14 are the NFL weekday schedule** (Sunday, Monday and Thursday games), not a hidden lattice.
6. **The fair test is the future.** Both rules make their only 2026 regular-season call on Oct 25, 2026 vs KC: each calls a loss vs KC if Seattle loses at DEN on Oct 15, 2026, and makes no call otherwise. Both rules are now in a dated ledger that scores them only on later games.

Bottom line. Nothing in Seattle's results since 1999 shows a calendar pattern that survives a neural network trained and tested on separate seasons, a shuffle test, or a correction for how many rules were tried. The two rules describe past games accurately, but their perfect-looking records are the kind that looking through many possible rules produces by chance. The honest way to settle it is to write rules down first and score them only on games that have not happened yet.

1 The question, the data and the tests

The idea being tested is that Seattle's wins and losses repeat or reverse at particular calendar intervals, a "lattice" built from spacings such as 3, 4, 7, 10, 11, 14, 21 and 28 days, and that particular combinations of past results predict the next one. Two specific rules came out of that search:

28/10 rule. If Seattle's results 28 days and 10 days before a game agree, predict that result again.

10-day, 4th-previous rule. If a game comes exactly 10 days after Seattle's previous game, and the previous and fourth-previous results agree, predict that result again.

Data

Every NFL game since 1999 comes from the nflverse `games.csv` file (compiled by Lee Sharpe): dates, scores, home and away, and closing point spreads. Seattle played 469 games in 1999–2025, playoffs included, and went 269–199–1 (57.5% wins). That win rate is the bar: a rule or model that cannot beat "Seattle wins" 57% of the time has not found anything. Ties count as neither result. Relocated franchises use their current codes (San Diego = LAC, St. Louis = LA, Oakland = LV).

Three tests, and one that is still running

1. **Out-of-sample (walk-forward).** Each model predicts a season using only earlier seasons: train on 1999–2003 and predict 2004, then train on 1999–2004 and predict 2005, and so on to 2025. A pattern that is real keeps working on games the model has never seen; a pattern that is noise does not.
2. **Shuffle test.** Keep every game's date, opponent and home/away, and every season's win–loss record, but deal each season's results out in random order. Repeat thousands of times. If a result also shows up in the shuffled histories, it cannot be evidence that the real order of results is special.
3. **Correction for searching.** When many rules are examined, some will look perfect by luck. The fair question is not "how unlikely is this rule's record?" but "how often does the best of all the rules examined look this good in a shuffled history?"
4. **Forward test.** A rule is written down with a date and scored only on games played afterward. This is the only test a rule found by searching old results can pass cleanly, and it is now set up (section 9).

2 The neural network

What it sees

The main network is given only the information the lattice idea says matters: Seattle's own results and when its games were played. For each game it receives 84 inputs:

- the last 8 results in game order (win = +1, loss = -1);
- the gaps in days between the last few games, and whether a gap spans an offseason;
- the result exactly d days earlier, for every d from 1 to 63 (0 when there was no game that day), which covers 28/10-style rules directly;
- the weekday of the game (Sunday, Monday, Thursday, Saturday, other).

For comparison, the same network was also trained on the last 8 results with no dates at all, on conventional information (point spread, home field, season record, point differential), and on everything together. Logistic regression is the same model with no hidden layer. The network has one hidden layer of 16 units, and each prediction averages 5 independently trained copies. Its settings were fixed before any test season was scored.

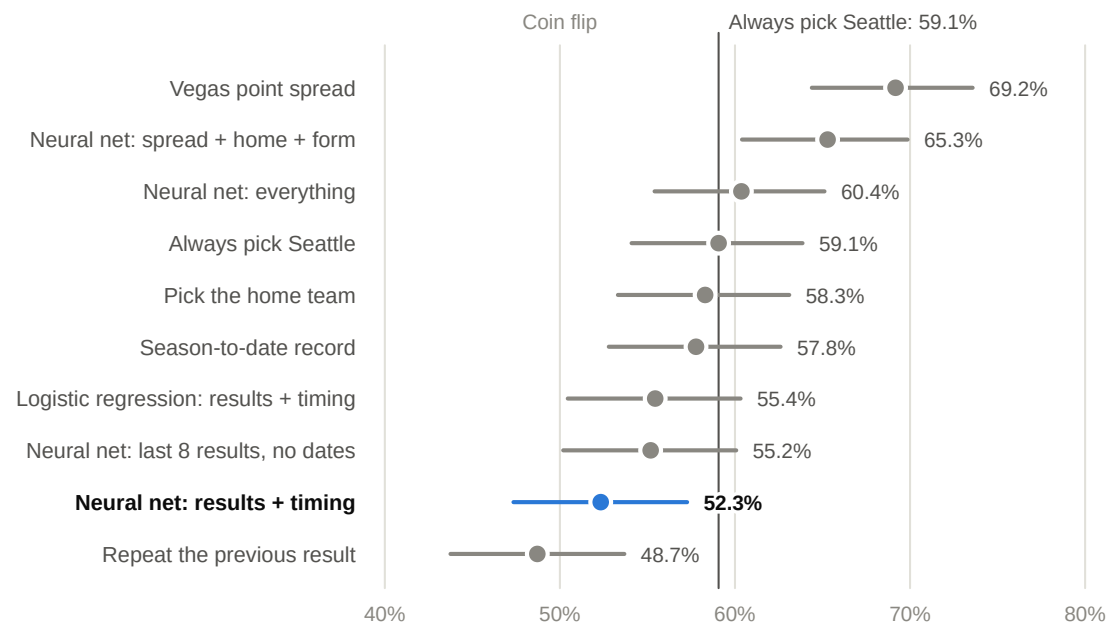


Figure 1. Share of 386 unseen games (2004–2025) each model picked correctly, with a 95% range. The highlighted network sees only results and timing. It lands below the line for always picking Seattle.

Table 1. Walk-forward results, 2004–2025. Log-loss rewards confident correct calls and punishes confident misses; lower is better, and always saying 50/50 scores 0.693.

Model	Games	Correct	95% range	Log-loss
Vegas point spread	386	69.2%	64%–74%	0.586
Neural net: spread + home + form	386	65.3%	60%–70%	0.610
Neural net: everything	386	60.4%	55%–65%	0.665
Always pick Seattle	386	59.1%	54%–64%	0.684
Pick the home team	386	58.3%	53%–63%	0.673
Season-to-date record	386	57.8%	53%–63%	0.685
Logistic regression: results + timing	386	55.4%	50%–60%	0.685
Neural net: last 8 results, no dates	386	55.2%	50%–60%	0.692
Neural net: results + timing	386	52.3%	47%–57%	0.695
Repeat the previous result	386	48.7%	44%–54%	0.698
28/10-day rule (in-sample)*	8	100.0%	68%–100%	–
10-day gap, 4th-previous rule (in-sample)*	10	90.0%	60%–98%	–

* The rules only make a call on 8 and 10 of these games and were found by looking at these same seasons, so their rows are not out-of-sample. Sections 4–6 correct them for the search.

What the results say

The results-and-timing network is worse than doing nothing clever. It and “always pick Seattle” disagreed on 130 games; the network was right on 52 of them and the simple rule on 78 (McNemar $p = 0.03$). Its log-loss (0.695) is worse than a constant guess (0.684), which is what a model fitting noise looks like.

Adding the timing inputs to the betting-line model hurts it. The spread + home + form network scored 65.3%; adding the results and timing inputs dropped it to 60.4% (log-loss 0.610 $\rightarrow$ 0.665). If the lattice carried information the market misses, this is where it would have shown up. Against the Vegas line itself, the everything network lost the disagreements 39 to 73 ($p = 0.002$).

It is not a matter of tuning. Smaller, larger and differently regularized versions of the results-and-timing network all land between 50.0% and 53.4%:

Table 2. The results + timing network under other settings (walk-forward, same games).

Setting	Correct	Log-loss
default (16 hidden, L2 0.01, early stopping)	52.3%	0.695
4 hidden units	52.3%	0.697
64 hidden units	52.6%	0.709
weak L2 (0.001)	50.0%	0.696
strong L2 (0.1)	53.4%	0.689
no early stopping, 200 epochs	50.0%	0.956

3 Shuffle test: is there any order in the sequence?

The walk-forward test asks whether the network can predict. The shuffle test asks a sharper question: does the *real order* of Seattle’s wins and losses contain anything that a random order of the same seasons does not? Each shuffle keeps the schedule and each season’s record and deals the results out at random; the whole walk-forward evaluation is then repeated on the shuffled history. This was done 200 times (with 3-network ensembles to keep it affordable).

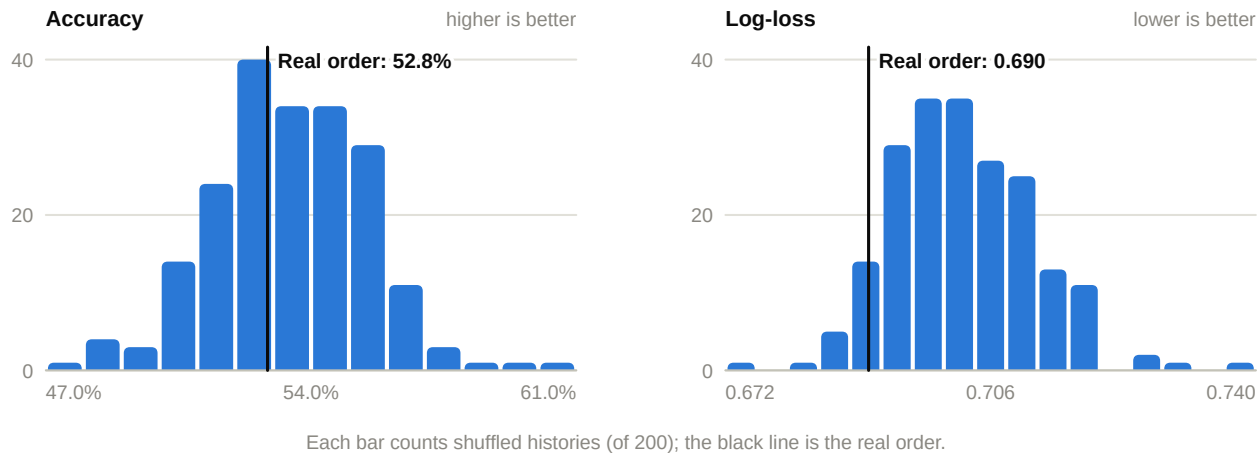


Figure 2. The results + timing network’s score on 200 shuffled histories (bars) and on the real history (black line). The real order sits inside the shuffled range on both measures: $p = 0.63$ for accuracy and 0.08 for log-loss.

Table 3. Real history against 200 shuffles for each model. p is the share of shuffles that did at least as well as the real order.

Model	Real	Shuffled (90% range)	p	Real log-loss	Shuffled	p
Neural net: results + timing	52.8%	50.2%–56.5%	0.63	0.690	0.703	0.08
Neural net: last 8 results, no dates	54.4%	50.2%–56.7%	0.44	0.696	0.700	0.35
Logistic regression: results + timing	54.9%	52.6%–58.6%	0.61	0.687	0.690	0.33
Repeat the previous result	48.7%	46.1%–56.5%	0.76	0.698	0.695	0.83

None of the rows is significant. The closest is the network's log-loss ($p = 0.08$); even if that edge were real, it would be too small to use, since the same network still loses to always picking Seattle. A control network that sees only the last 8 results, with no dates, behaves the same way ($p = 0.35$), so whatever small streakiness exists does not come from calendar spacing.

4 The 28/10 rule

Every Seattle game in 1999–2025 with a game exactly 28 days and exactly 10 days before it is listed below. The two ends of the 28-day span agree in 10 of 11 cases. The rule fires when the first two results agree, and the third matched all 8 times. Both counts match the hand count exactly.

Table 4. Every 28/10 sequence, 1999–2025.

Game	28 days before	10 days before	Result	Rule
Dec 7, 2008 vs NE	Nov 9, 2008 · L	Thu Nov 27, 2008 · L	L	✔ hit
Oct 28, 2012 at DET	Sep 30, 2012 · L	Thu Oct 18, 2012 · L	L	✔ hit
Dec 7, 2014 at PHI	Nov 9, 2014 · W	Thu Nov 27, 2014 · W	W	✔ hit
Nov 25, 2018 at CAR	Oct 28, 2018 · W	Thu Nov 15, 2018 · W	W	✔ hit
Oct 13, 2019 at CLE	Sep 15, 2019 · W	Thu Oct 3, 2019 · W	W	✔ hit
Oct 17, 2021 at PIT	Sep 19, 2021 · L	Thu Oct 7, 2021 · L	L	✔ hit
Dec 10, 2023 at SF	Nov 12, 2023 · W	Thu Nov 30, 2023 · L	L	no call
Oct 20, 2024 at ATL	Sep 22, 2024 · W	Thu Oct 10, 2024 · L	W	no call
Jan 5, 2025 at LA	Dec 8, 2024 · W	Thu Dec 26, 2024 · W	W	✔ hit
Oct 5, 2025 vs TB	Sep 7, 2025 · L	Thu Sep 25, 2025 · W	L	no call
Dec 28, 2025 at CAR	Nov 30, 2025 · W	Thu Dec 18, 2025 · W	W	✔ hit

Every “10 days before” game is a Thursday game. A 10-day gap only happens when a Thursday game is followed by a Sunday game, so the rule really says “after a Thursday game, compare it with four weeks earlier.” Seattle played 21 Thursday games in 1999–2025, the first on Dec 14, 2006, which is why the rule has so few cases.

On its own, 8 for 8 looks impressive. A coin would do it with probability 0.008. But “repeat the result” is not a coin flip: because good seasons stay good, this exact rule averages 61% hits even in shuffled seasons, and 8/8 still happens in 0.4% of them. The bigger issue is that 28/10 was not the only rule looked at, which is the subject of section 5.

Table 5. The strongest “if the games d1 and d2 days back agree, repeat it” rules in 1999–2025 (116 rules have at least 8 cases).

Rule	Cases	Repeated	Rate	p on its own	p after the search
49 & 28 days	67	45	67%	0.007	0.43
28 & 10 days	8	8	100%	0.008	0.57
35 & 10 days	8	8	100%	0.008	0.57
35 & 28 days	84	54	64%	0.01	0.62
42 & 13 days	10	9	90%	0.02	0.81
55 & 34 days	10	9	90%	0.02	0.81
39 & 18 days	9	8	89%	0.04	0.96
62 & 27 days	11	9	82%	0.07	0.99

Two other rules tie or beat 28/10 in the same data: 49 & 28 days (45/67) and 35 & 10 days (8/8). None of them survives the correction.

5 Why a search produces perfect records

Imagine 116 different rules, each with eight or more past cases. Even if none of them means anything, some will match their cases every time, the same way that among a hundred people flipping eight coins, someone will get eight heads. Finding one of them and then quoting its record is not the same as predicting that record in advance.

The correction below measures this directly. In each of 2,000 shuffled histories, every rule is scored and the most impressive one is kept. A real rule is only remarkable if it beats the best rule of most shuffled histories.

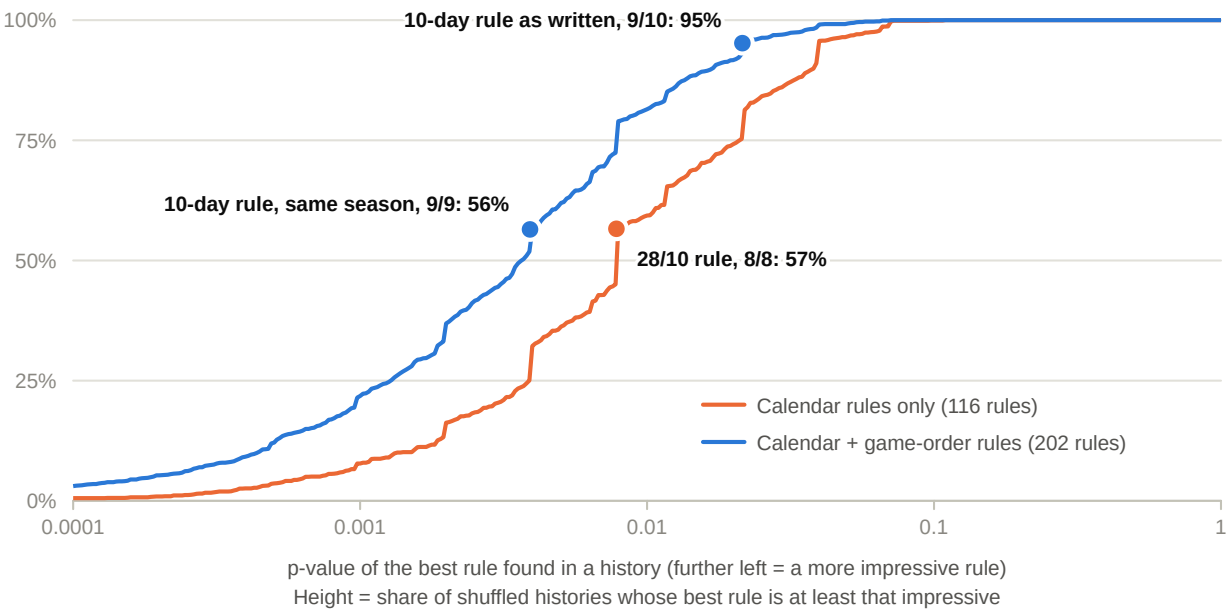


Figure 3. Each curve shows how often a shuffled history contains a best rule at least as impressive as the value on the horizontal axis. The dots are the real rules. The 28/10 rule is matched or beaten in 57% of shuffled histories when only calendar rules are searched; the best version of the 10-day rule in 56% when both families are searched; the 10-day rule as written in 95%.

How to read the corrected numbers. A value such as 56% means: in 56% of histories where the order of results is pure chance, searching the same rules finds one at least this good. Conventionally, a finding starts to count as evidence below 5%. These are far above it. The real history also contains 3 calendar rules at least as strong as 28/10, where shuffled histories average 1.0 and 10% of them have 3 or more.

These corrections are generous to the rules. They count only rules with at least 8 cases (the cutoff that includes 28/10), lookbacks of up to 63 days, and the two rule families defined here. They cannot account for rules that were adjusted after seeing a miss, as the 10-day rule was adjusted to avoid the 1996 case, so the true correction is larger.

6 The 10-day, 4th-previous rule

Because every 10-day gap is a Thursday followed by a Sunday, the 4th-previous game is usually the one exactly 28 days back, the same game the 28/10 rule uses. The two rules differ when a Monday game or a bye moves the calendar: the 2015 case spans 27 days (6 + 7 + 4 + 10) and counts only for the new rule.

As written, the rule is 9-for-10 in 1999–2025, not perfect. The miss is Sep 14, 2014 at LAC (L 21–30): the previous game was the Thu Sep 4, 2014 win, and the 4th-previous game was the Jan 11, 2014 playoff win, in the 2013 season. It becomes 9-for-9 only if all the games must be in one season, which the rule does not say. That condition is also the only way to reproduce a perfect 10-for-10 over 1976–2025: dropping playoff games does not remove this miss, because Seattle’s last regular-season game four back (Dec 15, 2013) was a win too.

One fact underneath every version

There were 14 games after a 10-day gap. Seattle repeated its Thursday result in 11 of them (79%); shuffled seasons give 55%, so this is unusual but not significant ($p = 0.10$). Every version of the rule is a subset of those 14 games. The grid shows which version calls which game: the perfect versions are the subsets that happen to leave out the 3 games that did not repeat.

Figure 4. Every Seattle game after a 10-day gap, 1999–2025, and each version of the rule:  hit  miss – no call (the two results disagree); blank: not eligible; shaded: the 4th-previous rule.

Game	Thursday	Result	Repeat?	k-th previous game, any season								k-th previous, same season only							
				2nd	3rd	4th	5th	6th	7th	8th	2nd	3rd	4th	5th	6th	7th	8th		
Dec 24, 2006 vs LAC	L	L	yes	✔	–	–	✔	–	–	✔	✔	✔	–	–	✔	–	–	✔	
Dec 7, 2008 vs NE	L	L	yes	✔	✔	✔	✔	–	✔	✔	✔	✔	✔	✔	✔	–	✔	✔	
Oct 28, 2012 at DET	L	L	yes	–	–	✔	–	–	✔	✔			✔	–	–	✔			
Sep 14, 2014 at LAC	W	L	no	✖	✖	✖	✖	–	✖	–									
Dec 7, 2014 at PHI	W	W	yes	✔	–	✔	✔	✔	–	–	✔	–	✔	✔	✔	–	–		
Nov 1, 2015 at DAL	W	W	yes	–	–	✔	✔	–	–	–	–	–	✔	✔	–	–			
Nov 25, 2018 at CAR	W	W	yes	–	–	✔	✔	–	✔	✔	–	–	✔	✔	–	✔	✔		
Oct 13, 2019 at CLE	W	W	yes	✔	–	✔	✔	–	✔	✔	✔	–	✔	✔					
Oct 17, 2021 at PIT	L	L	yes	–	✔	✔	–	✔	–	–	–	✔	✔	–					
Dec 10, 2023 at SF	L	L	yes	✔	✔	–	✔	–	–	✔	✔	✔	–	✔	–	–	✔		
Oct 20, 2024 at ATL	L	W	no	✖	✖	–	–	–	–	✖	✖	✖	–	–	–				
Jan 5, 2025 at LA	W	W	yes	–	–	✔	✔	✔	✔	–	–	–	✔	✔	✔	✔	–		
Oct 5, 2025 vs TB	W	L	no	✖	✖	–	✖	✖	–	–	✖	✖	–						
Dec 28, 2025 at CAR	W	W	yes	✔	✔	✔	✔	–	✔	✔	✔	✔	✔	✔	–	✔	✔		
Hits / calls				6/9	4/7	9/10	9/11	3/4	6/7	7/8	6/8	4/6	9/9	9/9	2/2	5/5	5/5		

The 4th-previous version is 9/10 as written and 9/9 same-season only. It is not the only perfect-looking choice: the 5th-previous same-season version is 9/9, and the 7th and 8th are 5/5 and 5/5.

Corrected for the search

86 game-order rules have at least 8 cases (every gap up to 21 days or any gap, positions 2 to 8, any season or same season only). With the 116 calendar rules that makes 202.

Table 6. The 10-day, 4th-previous rule, measured four ways.

Version	Record	p on its own	vs shuffled seasons	after the game-order search	after both searches
Same season only	9/9	0.004	< 0.001	0.39	0.56
As written	9/10	0.02	0.01	0.78	0.95

Table 7. The strongest game-order rules found, corrected for both searches.

Rule	Cases	Repeated	Rate	p on its own	p after both searches
10-day gap, 4th previous, same season	9	9	100%	0.004	0.56
10-day gap, 5th previous, same season	9	9	100%	0.004	0.56
10-day gap, 4th previous	10	9	90%	0.02	0.95
6-day gap, 2nd previous	23	6	26%	0.03	0.98
8-day gap, 8th previous, same season	9	8	89%	0.04	0.99
14-day gap, 5th previous	20	5	25%	0.04	0.99

Other teams. For the other 31 teams the same rule is right 173 of 289 times (60%) as written and 145 of 241 (60%) same-season only, against 57% and 57% in their shuffled seasons. Across the league it is a “good seasons stay good” rule and nothing more. The neural network in section 2 had exactly the inputs this rule needs (the last 8 results in order and the gap before each game); with 10 cases in 27 seasons it cannot be told apart from the other patterns noise produces.

7 Intervals, repeats and reversals

Where 3, 4, 7, 10, 11 and 14 come from

Nearly every NFL game is played on a Sunday, Monday, Thursday or Saturday, so the gap between two games is a week plus or minus a weekday shift. The chart counts every gap between back-to-back Seattle games in 1999–2025.

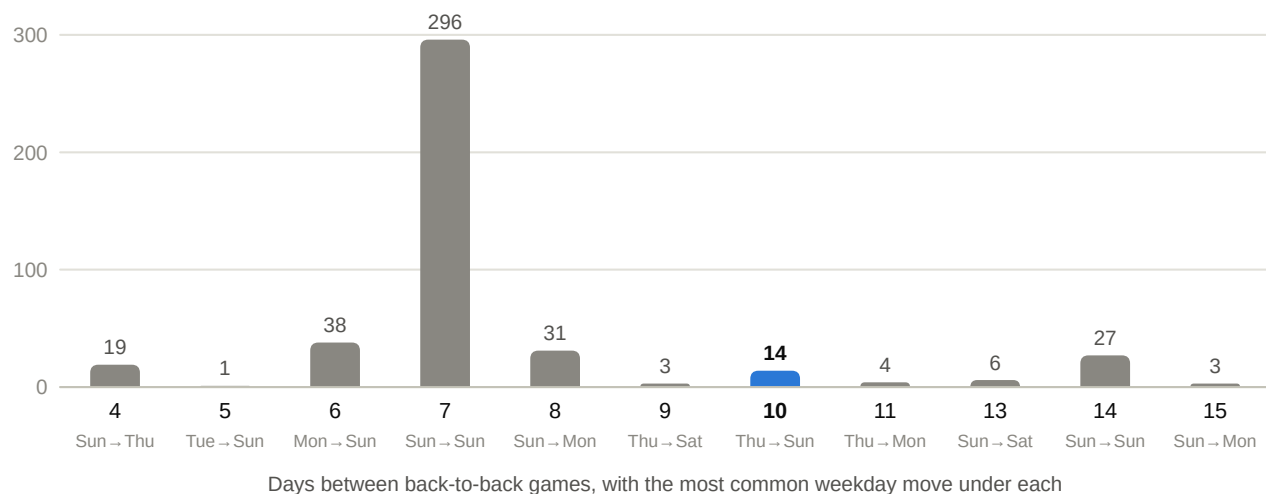


Figure 5. Gaps between back-to-back games. Almost all are 7 days (Sunday to Sunday); 4 and 10 are the two halves of a Thursday game (Sunday → Thursday, Thursday → Sunday), 8 and 6 the two halves of a Monday game, 14 a bye week.

So $14 = 7 + 7 = 8 + 6 = 4 + 10$ is the same two-week span with the middle game moved to a Monday or a Thursday; $27 = 6 + 7 + 4 + 10$ is four weeks that start with a Monday game; 21 is three weeks; and 11 is a Thursday game followed by a Monday game. These numbers appear because of how the league schedules games, which is why they recur in every team's history.

Do results repeat or reverse at particular gaps?

Table 8. Back-to-back games by the days between them. “Same” means the second game repeated the first result; “Shuffled” is what reshuffled seasons give. q corrects p for the number of gaps checked (false-discovery rate).

Gap	Weekdays	Pairs	Same	Rate	Shuffled	p	q
4 days	Sun → Thu	19	10	53%	55%	0.98	1.00
6 days	Mon → Sun, Sun → Sat	38	18	47%	52%	0.63	0.88
7 days	Sun → Sun, Sat → Sat	294	142	48%	52%	0.16	0.37
8 days	Sun → Mon, Sat → Sun	31	18	58%	53%	0.60	0.88
10 days	Thu → Sun	14	11	79%	55%	0.10	0.34
13 days	Sun → Sat, Mon → Sun	6	4	67%	62%	1.00	1.00
14 days	Sun → Sun, Sat → Sat	27	9	33%	54%	0.05	0.34

Table 9. Every pair of games up to 400 days apart (not only back-to-back): the eight most extreme of 87 intervals.

Interval	Pairs	Same	Rate	Shuffled	p	q
14 days	276	124	45%	52%	0.006	0.41
55 days	31	24	77%	54%	0.009	0.41
351 days	43	30	70%	51%	0.02	0.51
105 days	60	24	40%	53%	0.05	0.83
302 days	27	9	33%	52%	0.05	0.83
288 days	24	8	33%	52%	0.06	0.83
18 days	20	15	75%	54%	0.07	0.83
266 days	73	46	63%	53%	0.08	0.83

A few intervals have raw p-values under 0.05, as expected when 87 are checked; none survives the correction (lowest q = 0.41, and 0.34 for back-to-back games). The most extreme, 14-day pairs repeating less often than usual, would be a reversal pattern if it were real; it is within what 87 tries produce.

8 All 32 teams

If Seattle had a hidden lattice, the same search on other teams should come up comparatively empty. It does not. 26 of the other 31 teams have a calendar rule that looks at least as strong as 28/10 by the same measure, spread over 24 different pairs of intervals, and 11 of them have at least one perfect rule with 8+ cases. After correcting for the search, 2 of 32 teams' best rules pass at the 5% level, where chance alone gives about 1.6 (PHI: 35 & 21 days, 56/74; SF: 42 & 28 days, 62/83).

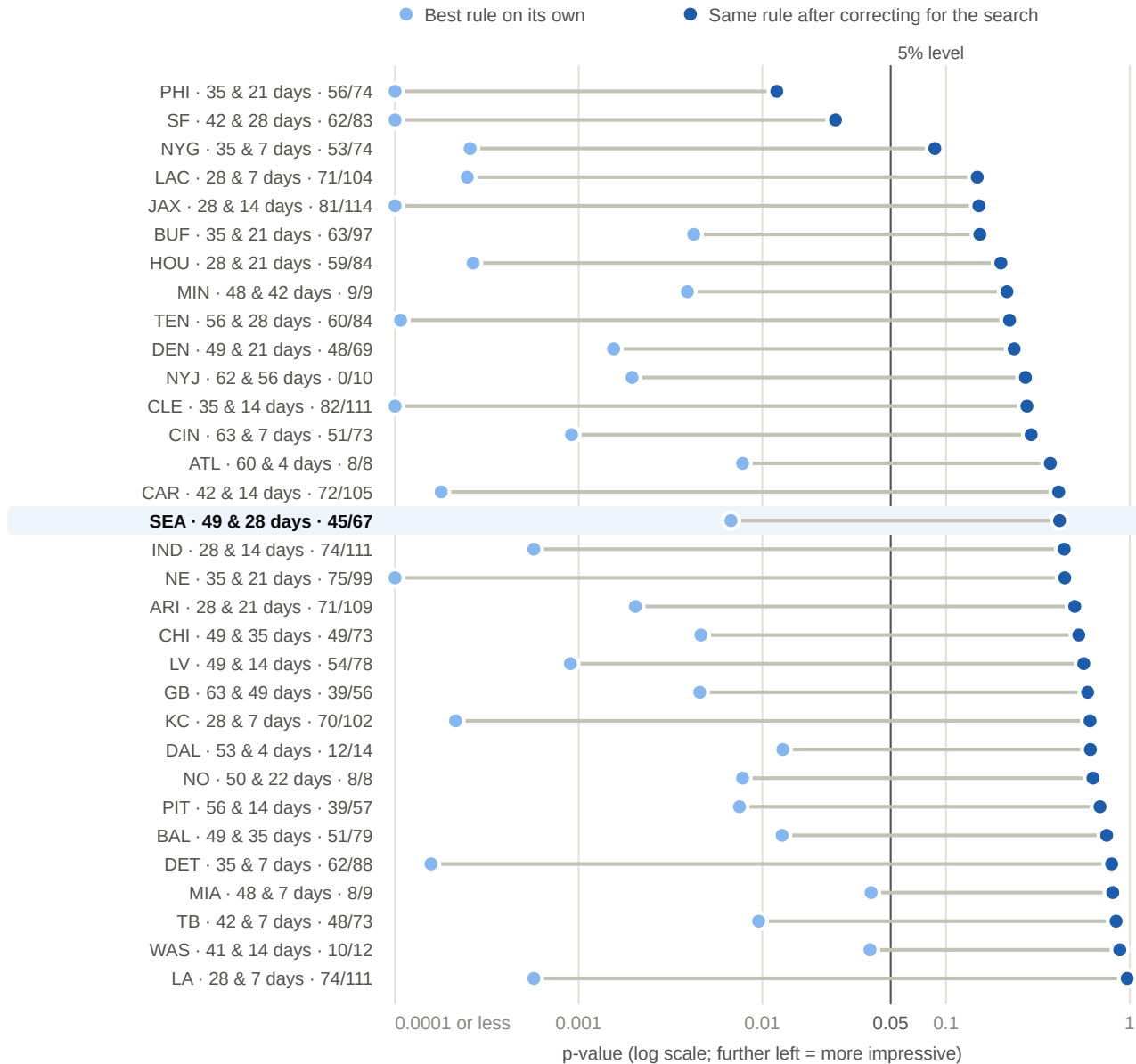


Figure 6. Each team's most impressive calendar rule (1999–2025): its p-value on its own, against a coin flip (light), and after correcting for the search (dark). Every team has a rule that looks remarkable on its own; almost none survives the correction. “A different pattern for every team” is what noise produces.

Table 10. Per team, 1999–2025: both named rules, the best calendar rule, and the results + timing network (walk-forward, 3-network ensembles) against always predicting the team's more common result.

Team	28/10	10-day, 4th prev.	Best calendar rule	p after search	Network	Base rate
PHI	4/6	9/13	35 & 21 days: 56/74	0.01	57.1%	58.1%
SF	4/6	5/7	42 & 28 days: 62/83	0.02	56.3%	50.8%
NYG	2/5	4/8	35 & 7 days: 53/74	0.09	52.4%	53.2%
LAC	4/5	5/6	28 & 7 days: 71/104	0.15	53.0%	43.0%
JAX	6/7	6/8	28 & 14 days: 81/114	0.15	60.4%	57.9%
BUF	0/1	1/2	35 & 21 days: 63/97	0.15	51.3%	48.1%
HOU	4/6	6/10	28 & 21 days: 59/84	0.20	51.6%	53.8%
MIN	4/8	5/10	48 & 42 days: 9/9	0.21	53.5%	48.9%
TEN	6/10	6/12	56 & 28 days: 60/84	0.22	53.8%	44.3%
DEN	5/9	8/13	49 & 21 days: 48/69	0.23	53.9%	53.9%
NYJ	5/6	8/13	62 & 56 days: 0/10	0.27	54.9%	53.3%
CLE	4/6	4/7	35 & 14 days: 82/111	0.28	64.9%	66.3%
CIN	1/4	4/8	63 & 7 days: 51/73	0.29	52.2%	50.8%
ATL	2/6	2/6	60 & 4 days: 8/8	0.37	50.7%	45.5%
CAR	3/5	5/8	42 & 14 days: 72/105	0.41	53.3%	47.8%
SEA	8/8	9/10	49 & 28 days: 45/67	0.41	52.8%	59.1%
IND	4/6	7/10	28 & 14 days: 74/111	0.44	59.3%	57.9%
NE	6/6	11/12	35 & 21 days: 75/99	0.44	69.0%	68.5%
ARI	2/4	1/3	28 & 21 days: 71/109	0.50	54.2%	55.9%
CHI	4/8	3/9	49 & 35 days: 49/73	0.53	49.0%	51.5%
LV	3/4	4/7	49 & 14 days: 54/78	0.56	57.1%	60.7%
GB	6/8	7/12	63 & 49 days: 39/56	0.59	59.2%	60.2%
KC	3/6	9/13	28 & 7 days: 70/102	0.61	62.9%	54.5%
DAL	6/13	8/15	53 & 4 days: 12/14	0.61	51.8%	46.1%
NO	2/6	3/8	50 & 22 days: 8/8	0.63	51.3%	52.4%
PIT	4/8	9/13	56 & 14 days: 39/57	0.69	60.9%	63.0%
BAL	3/5	7/10	49 & 35 days: 51/79	0.75	55.0%	59.2%
DET	11/16	11/18	35 & 7 days: 62/88	0.80	61.6%	58.6%
MIA	1/6	3/9	48 & 7 days: 8/9	0.81	48.2%	47.9%
TB	3/4	5/7	42 & 7 days: 48/73	0.84	51.8%	49.9%
WAS	3/4	3/6	41 & 14 days: 10/12	0.88	54.4%	58.8%
LA	3/4	4/6	28 & 7 days: 74/111	0.97	50.3%	43.6%

A team-specific lattice would show up in each team's own network. Trained on one team at a time, the results-and-timing network beat its team's base rate on accuracy for 18 of 32 teams but on log-loss for only 9 of 32. Losing to the base rate for most teams (sign test $p = 0.02$) is the signature of fitting noise. Pooled across the other 31 teams, the 28/10 rule is right 118 of 198 times (60%), against 57% in their shuffled seasons.

9 What happens next

Next game: Sun Oct 4, 2026 vs LAC

Seattle is favored by 6.5 points. Every model was retrained on all 471 finished games through Sep 27, 2026.

Table 11. Probability that Seattle wins on Oct 4, 2026.

Model	Win probability	Out-of-sample record (section 2)
Always pick Seattle (base rate)	58%	59.1%
Vegas point spread	76%	69.2%
Neural net: spread + home + form	72%	65.3%
Neural net: everything	80%	60.4%
Neural net: results + timing	66%	52.3%
Neural net: last 8 results, no dates	71%	55.2%

Only the models that use the point spread beat the base rate out of sample, so the Vegas-based figure is the one to trust. The results-and-timing figure is shown for completeness; its track record says it is noise.

The live test: Oct 25, 2026 vs KC

Both rules compare the same two games: Sun Sep 27, 2026 at WAS (L) and Thu Oct 15, 2026 at DEN (not played yet). **If Seattle loses at DEN on Oct 15, 2026, both rules call a loss vs KC. If not, neither makes a call.** This is one shared test, not two: the rules cannot disagree on this game.

The ledger

`ledger.json` records each rule with the date it was written down, and `node cli.js ledger` scores it only on games played after that date. New rules should be added the day they are found; their record from then on is the only one that counts as evidence.

Rule	Written down	Forward record	Next call
If Seattle's results 28 and 10 days before a game agree, predict that result again.	Sep 28, 2026	0/0	Oct 25, 2026 vs KC: calls a loss if Seattle loses at DEN on Oct 15, 2026, and makes no call otherwise
If Seattle's game is exactly 10 days after its previous game, and the previous and fourth-previous results agree, predict that result again.	Sep 28, 2026	0/0	Oct 25, 2026 vs KC: calls a loss if Seattle loses at DEN on Oct 15, 2026, and makes no call otherwise

How long until the ledger can decide?

These rules fire rarely. Since 2012, when Thursday games became a weekly fixture, they have made about 0.6 calls a season. To tell a rule that is truly right 90% of the time from the 57% that “good seasons stay good” already produces, a fair test needs:

Table 12. Forward calls needed. “Hits needed” is the fewest hits that would be convincing (a one-sided exact test at the 5% level against 57%); “chance of passing” assumes the rule really is right 90% of the time.

Forward calls	Hits needed	False-alarm rate	Chance of passing if the rule is real	Seasons at the current rate
5	not possible	–	0%	about 9
10	9 of 10	3.1%	74%	about 18
15	13 of 15	1.6%	82%	about 26
20	16 of 20	2.9%	96%	about 35
30	22 of 30	5.0%	100%	about 53

Even a perfect run of 5 forward calls would not be convincing, and reaching 15 calls at the current rate takes roughly 26 seasons. Until then, the rules should be treated as unproven, and results from the rest of the league (where the same rules already run at about 60%) are the faster check.

10 Conclusions

What the evidence supports

- Seattle’s results since 1999 are well described by team strength (season records, home field, the point spread). Once that is known, the calendar and the order of past results add nothing a network can use; adding them makes predictions worse.
- The 28/10 and 10-day rules are accurate descriptions of past games. Their records are what a search of this size produces by chance, and their success above 50% comes from good seasons staying good, which holds for every team.
- The recurring numbers (3, 4, 7, 10, 11, 14, 21, 27, 28) come from the NFL’s weekday schedule.

What would change this conclusion

- A rule written down in advance that keeps hitting on future games well above 57%, over enough calls (section 9).
- A rule that works across many teams at once, beyond what their shuffled seasons give. That would be testable quickly, because the league plays about 32 times as many games.
- A model that uses the lattice inputs and beats the point spread out of sample. None of the models here comes close.

A note on the other numbers

The year arithmetic ($1999 - 1976 = 23$; $2160 - 1976 = 184 = 8 \times 23$; $2160 - 1999 = 161 = 7 \times 23$) is one coincidence, not two: once $2160 - 1976$ is a multiple of 23, subtracting $1999 - 1976 = 23$ makes $2160 - 1999$ one automatically. Numbers such as 1764, 1998, 2760 and 1380 could not be tested without a definition of what they count. Any rule that can be stated in days or game positions can be checked with `node cli.js query` and registered in the ledger.

A Method details

- **Data.** nflverse `games.csv`, 469 Seattle games in 1999–2025 (468 with a winner). Features for a game use only earlier results plus facts known before kickoff (date, weekday, home/away, spread); a unit test flips every later

result and checks that a game's inputs do not change.

- **Network.** Inputs → 16 tanh units → sigmoid output. Full-batch Adam (learning rate 0.01), L2 penalty 0.01, early stopping on a random 20% of the training games (patience 30, at most 400 epochs), average of 5 seeds. Logistic regression is the same code with no hidden layer. Written without dependencies in Node.js.
- **Walk-forward.** Expanding window: train 1999..Y-1, predict every game of season Y, for Y = 2004..2025 (386 games).
- **Shuffle null.** Within each season, results are permuted across that season's games (regular season and playoffs), keeping dates, opponents, home/away and the season record.
- **Rule families.** Calendar: games exactly d1 and d2 days before agree ($d2 < d1 \leq 63$). Game-order: the gap before the game is g days (every $g \leq 21$, or any gap) and the previous and k-th previous results agree ($k = 2..8$), any season or same season only. Rules need at least 8 calls.
- **Corrections.** “p after the search” is the share of 2,000 shuffled histories whose best rule (smallest two-sided binomial p) is at least as small (min-p family-wise correction). Interval tables use Benjamini–Hochberg q-values; accuracy ranges are Wilson 95% intervals; paired model comparisons use exact McNemar tests.

B Reproducing this report

```
cd seahawks-nn
npm test                # unit tests
node cli.js report      # full analysis (about 4 minutes)
node cli.js pdf         # this PDF, from results/results.json
node cli.js query --gap 10 --position 4 --same-season
node cli.js ledger      # forward record of the registered rules
npm run update-data     # new games from nflverse (needs GitHub access)
```

C Glossary

Out-of-sample / walk-forward

Scoring a model only on games it was not trained on, season by season, in the order they happened.

p-value

How often chance alone would produce a result at least this extreme. Small is surprising; 0.05 (5%) is the conventional line.

p after the search (family-wise)

The p-value corrected for having looked at many rules: how often the best rule in a chance-only history looks this good.

Shuffle test

Re-running an analysis on histories where each season's results are put in random order, to see what chance produces with the same schedules and records.

Log-loss

A score for probability forecasts that punishes confident misses. Lower is better; always saying 50% scores 0.693.

Base rate

The accuracy of always predicting the more common result (for Seattle, a win).